

Homework (Habanero)

Q1. Rationalize the denominator of $\frac{2}{\sqrt{3}}$.

- (A) $\frac{2\sqrt{3}}{3}$
- (B) $\frac{2}{3\sqrt{3}}$
- (C) $\frac{2\sqrt{2}}{3}$
- (D) $\frac{2}{\sqrt{2}}$
- (E) $\frac{2\sqrt{3}}{9}$

Q2. Simplify $\frac{1}{\sqrt{12}}$ by rationalizing the denominator.

- (A) $\frac{\sqrt{3}}{6}$
- (B) $\frac{\sqrt{2}}{4}$
- (C) $\frac{\sqrt{12}}{12}$
- (D) $\frac{1}{2\sqrt{3}}$
- (E) $\frac{\sqrt{3}}{12}$

Q3. The dosage of a certain medicine is $\frac{5}{\sqrt{d}}$ mg per day, where d is the number of days since the start of treatment. Rationalize the denominator to find the dosage after $d = 4$ days.

- (A) $\frac{5\sqrt{2}}{4}$
- (B) $\frac{5}{2}$
- (C) $\frac{5\sqrt{4}}{4}$
- (D) $\frac{5}{\sqrt{2}}$
- (E) $\frac{5\sqrt{2}}{2}$

Q4. Rationalize the denominator of $\frac{3}{2+\sqrt{5}}$.

- (A) $\frac{3(2-\sqrt{5})}{-1}$
- (B) $\frac{3(2+\sqrt{5})}{4+5}$
- (C) $\frac{3(2-\sqrt{5})}{4-5}$
- (D) $\frac{3(2+\sqrt{5})}{-1}$
- (E) $\frac{3(2-\sqrt{5})}{9}$

Q5. Simplify $\frac{2}{\sqrt{8}+\sqrt{2}}$ by rationalizing the denominator.

- (A) $\frac{2(\sqrt{8}-\sqrt{2})}{6}$
- (B) $\frac{2(\sqrt{8}+\sqrt{2})}{8}$
- (C) $\frac{2(\sqrt{8}-\sqrt{2})}{6}$
- (D) $\frac{2(\sqrt{2}-\sqrt{8})}{-6}$
- (E) $\frac{2(\sqrt{8}+\sqrt{2})}{6}$

Q6. The population of a certain cell grows according to the function $P(t) = \frac{1}{\sqrt{t+1}}$. Rationalize the denominator to find the population after $t = 3$ days.

- (A) $\frac{\sqrt{4}}{4}$
- (B) $\frac{1}{2}$
- (C) $\frac{1}{\sqrt{2}}$
- (D) $\frac{\sqrt{4}}{2}$
- (E) $\frac{1}{4}$

Q7. Simplify $\frac{3}{\sqrt{18}-\sqrt{2}}$ by rationalizing the denominator.

- (A) $\frac{3(\sqrt{18}+\sqrt{2})}{16}$
- (B) $\frac{3(\sqrt{2}-\sqrt{18})}{16}$
- (C) $\frac{3(\sqrt{18}-\sqrt{2})}{16}$
- (D) $\frac{3(\sqrt{2}+\sqrt{18})}{18}$
- (E) $\frac{3(\sqrt{18}+\sqrt{2})}{18}$

Q8. Rationalize the denominator of $\frac{2}{\sqrt{20}-\sqrt{5}}$.

- (A) $\frac{2(\sqrt{20}+\sqrt{5})}{15}$
- (B) $\frac{2(\sqrt{5}+\sqrt{20})}{15}$
- (C) $\frac{2(\sqrt{20}-\sqrt{5})}{15}$
- (D) $\frac{2(\sqrt{5}-\sqrt{20})}{15}$
- (E) $\frac{2(\sqrt{20}+\sqrt{5})}{5}$

Open Ended Questions (FRQ)

Q9. The dosage of a certain medicine is $\frac{10}{\sqrt{t}}$ mg per day, where t is the number of hours since the start of treatment. Rationalize the denominator and find the dosage after $t = 9$ hours. Show all work and provide a final numerical answer.

Q10. The population of a certain bacteria colony grows according to the function $P(t) = \frac{1}{\sqrt{t^2+1}}$. Rationalize the denominator and find the population after $t = 2$ days. Show all work and provide a final numerical answer.

Chef's Tips

- Provide step-by-step solutions for each problem.
- Emphasize the importance of rationalizing the denominator in simplifying radical expressions.
- Encourage students to check their work by plugging in values for variables.